

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

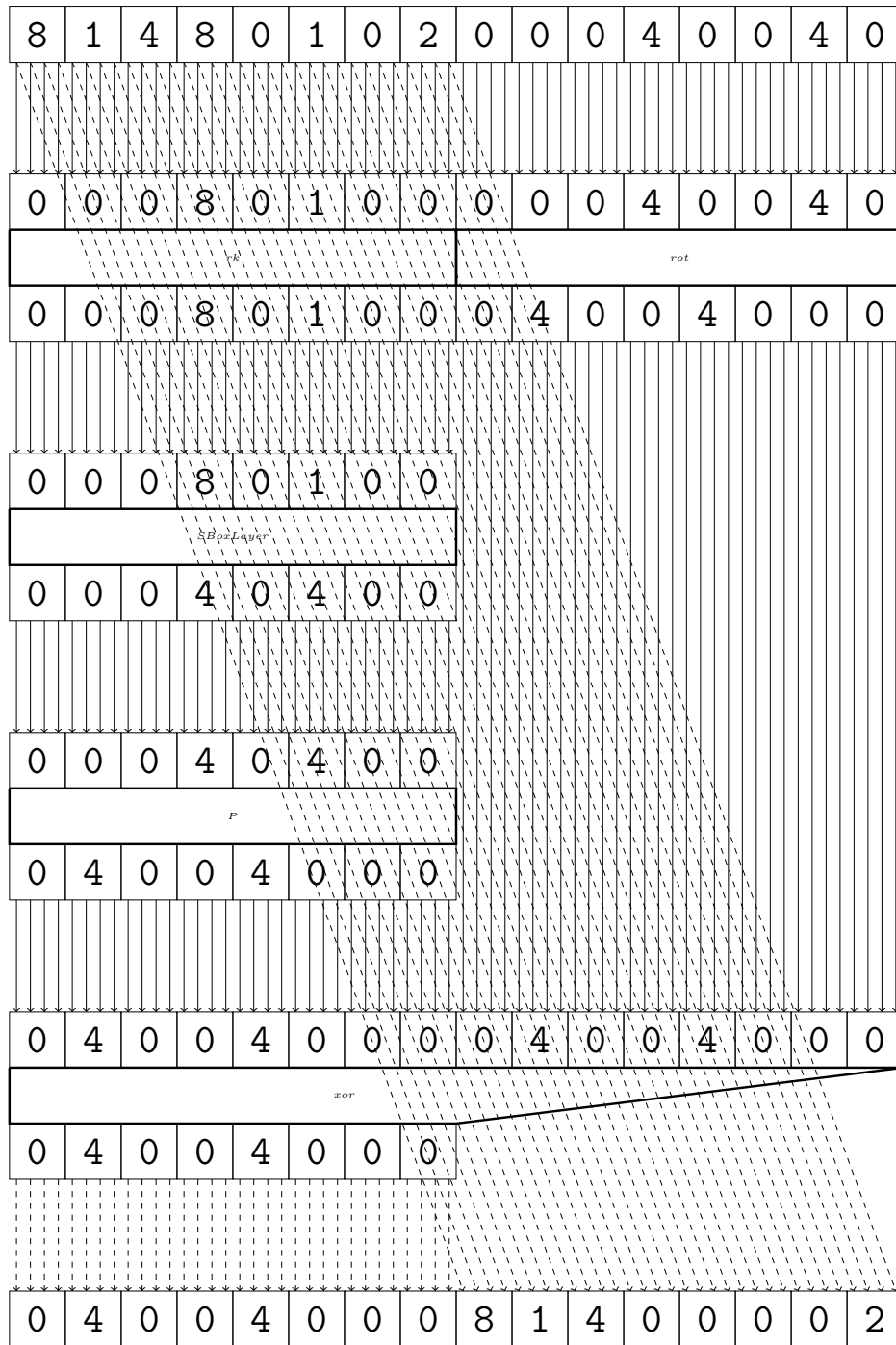
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-24}

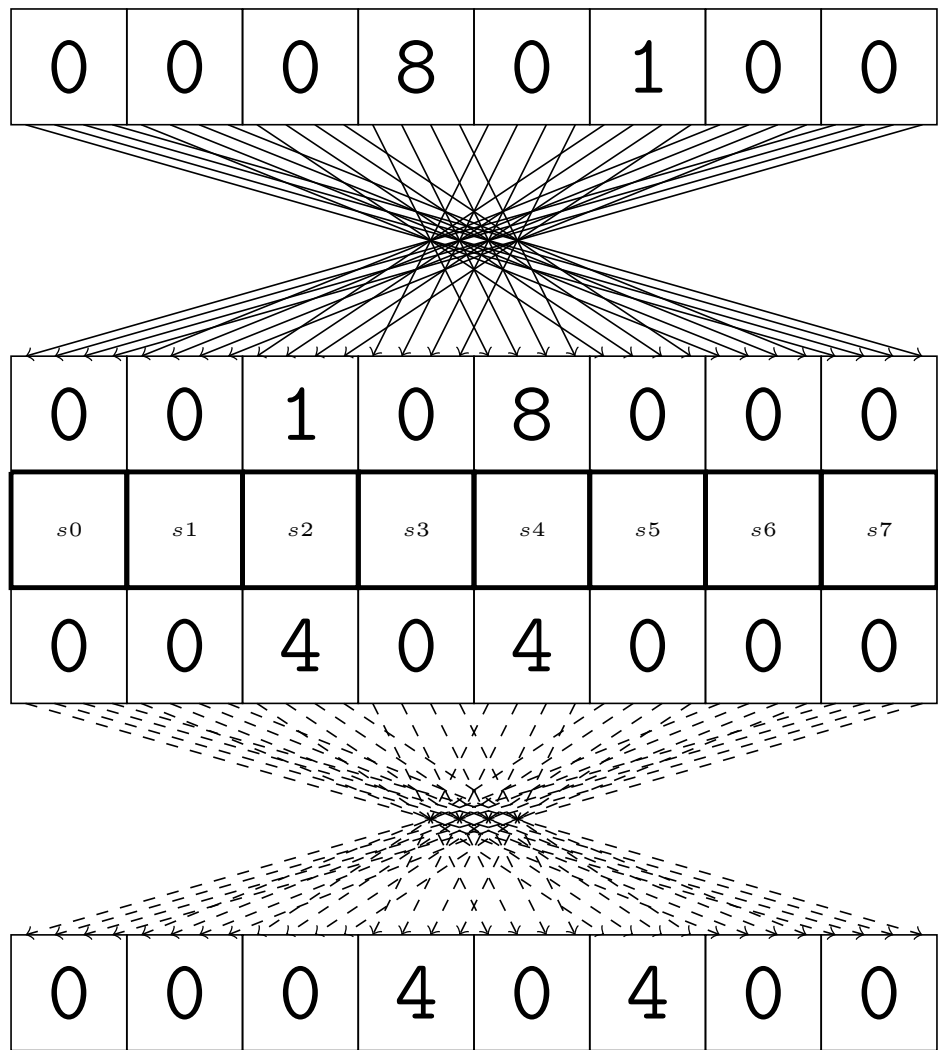
8	1	4	8	0	1	0	2	0	0	0	4	0	0	4	0
8	1	4	8	0	1	0	2	0	0	0	4	0	0	4	0
LBlockRound															
0	4	0	0	4	0	0	0	8	1	4	0	0	0	0	2
0	4	0	0	4	0	0	0	8	1	4	0	0	0	0	2
LBlockRound															
4	0	0	0	0	2	8	1	0	0	0	0	0	a	9	
4	0	0	0	0	2	8	1	0	0	0	0	0	a	9	
LBlockRound															
0	0	0	0	a	9	0	0	4	0	0	0	0	0	8	0
0	0	0	0	a	9	0	0	4	0	0	0	0	0	8	0
LBlockRound															
0	0	0	0	8	0	4	0	0	0	0	0	0	0	0	0
0	0	0	0	8	0	4	0	0	0	0	0	0	0	0	0
LBlockRound															
0	0	0	0	0	0	0	0	0	0	0	0	8	0	4	0
0	0	0	0	0	0	0	0	0	0	0	0	8	0	4	0
LBlockRound															
0	0	8	0	4	0	0	0	8	0	0	0	0	8	0	0
0	0	8	0	4	0	0	0	8	0	0	0	0	8	0	0
LBlockRound															
0	0	0	8	0	0	8	0	0	0	0	0	0	0	0	0
0	0	0	8	0	0	8	0	0	0	0	0	0	0	0	0
LBlockRound															
0	0	0	0	0	0	0	0	0	0	0	8	0	0	8	0
0	0	0	0	0	0	0	0	0	0	0	8	0	0	8	0
LBlockRound															
0	8	0	0	8	0	0	0	0	0	0	d	0	9	0	0
0	8	0	0	8	0	0	0	0	0	0	d	0	9	0	0
LBlockRound															
0	d	0	9	0	0	0	0	0	8	1	8	8	0	0	0
0	d	0	9	0	0	0	0	0	8	1	8	8	0	0	0
LBlockRound															
1	8	8	0	0	0	0	8	8	0	0	0	0	0	8	0
1	8	8	0	0	0	0	8	8	0	0	0	0	0	8	0
LBlockRound															
0	0	0	0	8	0	8	0	1	8	8	0	4	d	0	8
0	0	0	0	8	0	8	0	1	8	8	0	4	d	0	8
LBlockRound															
1	8	8	0	4	d	0	8	0	0	0	0	8	0	8	0
1	8	8	0	4	d	0	8	0	0	0	0	8	0	8	0

2 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

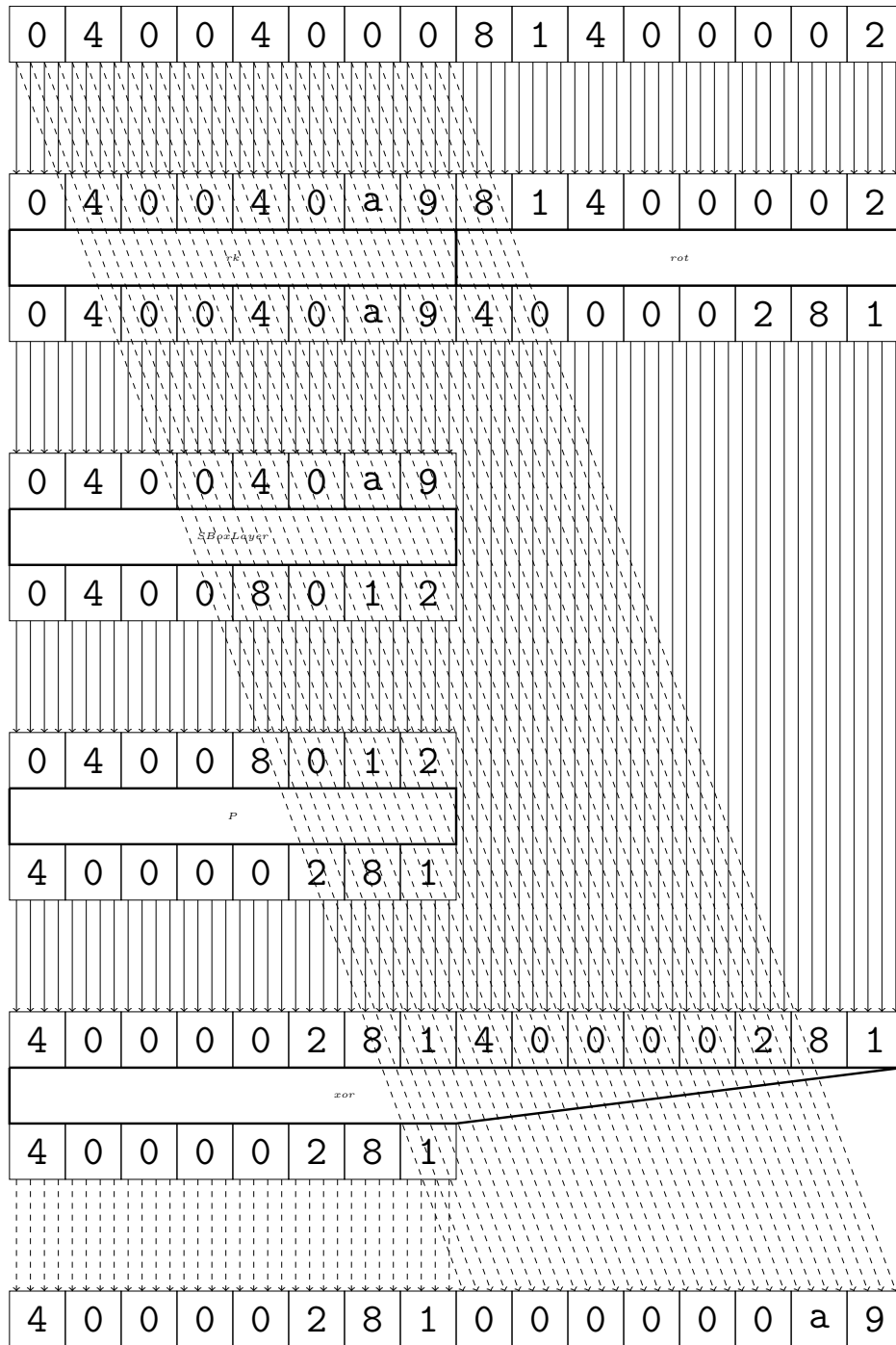
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



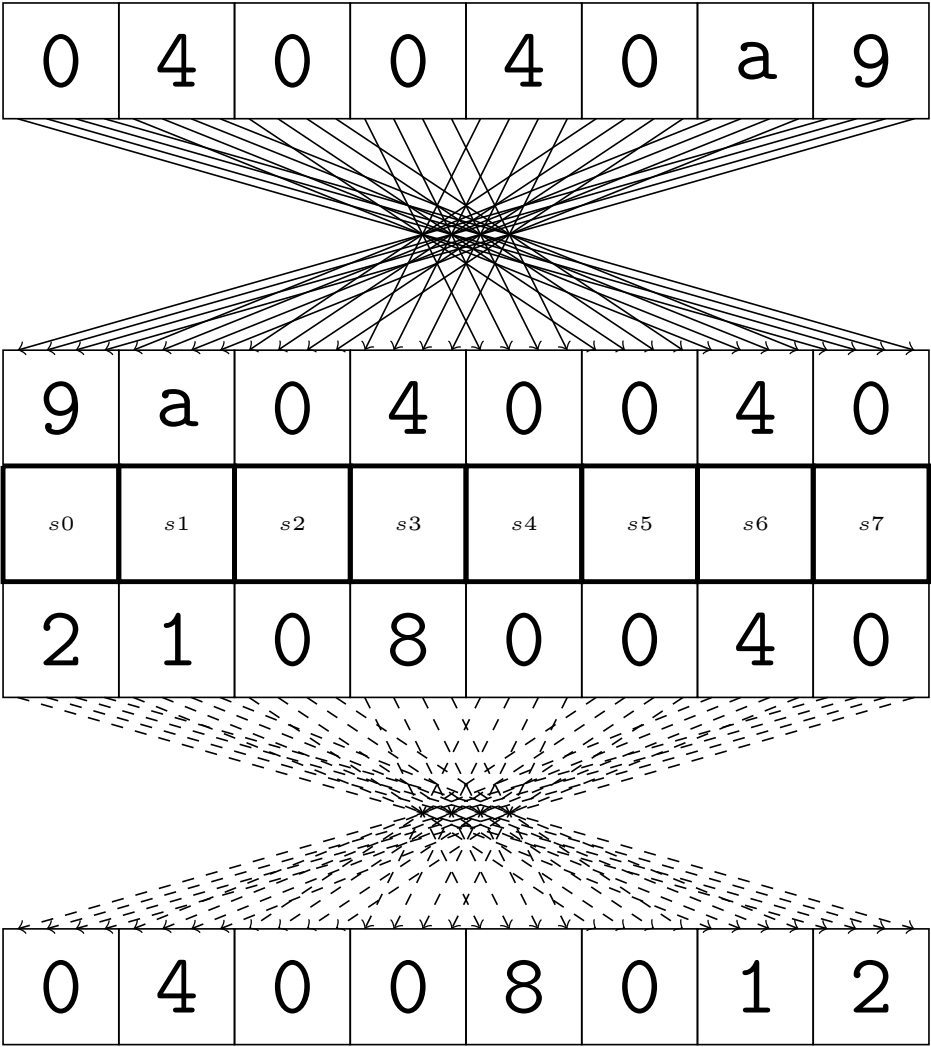
4 LBlock_round

Maximal linear correlation: $2^{-4.0}$



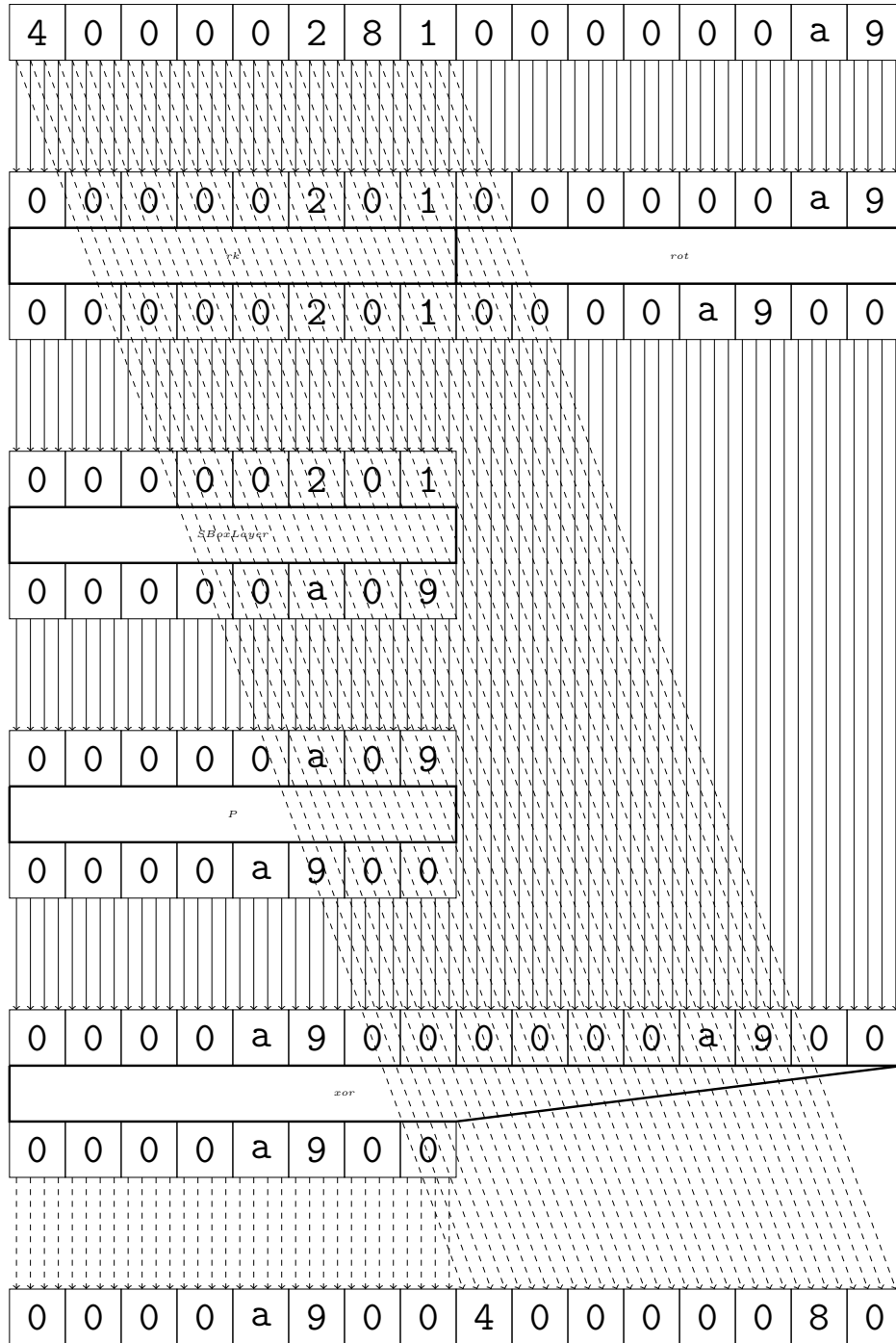
5 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



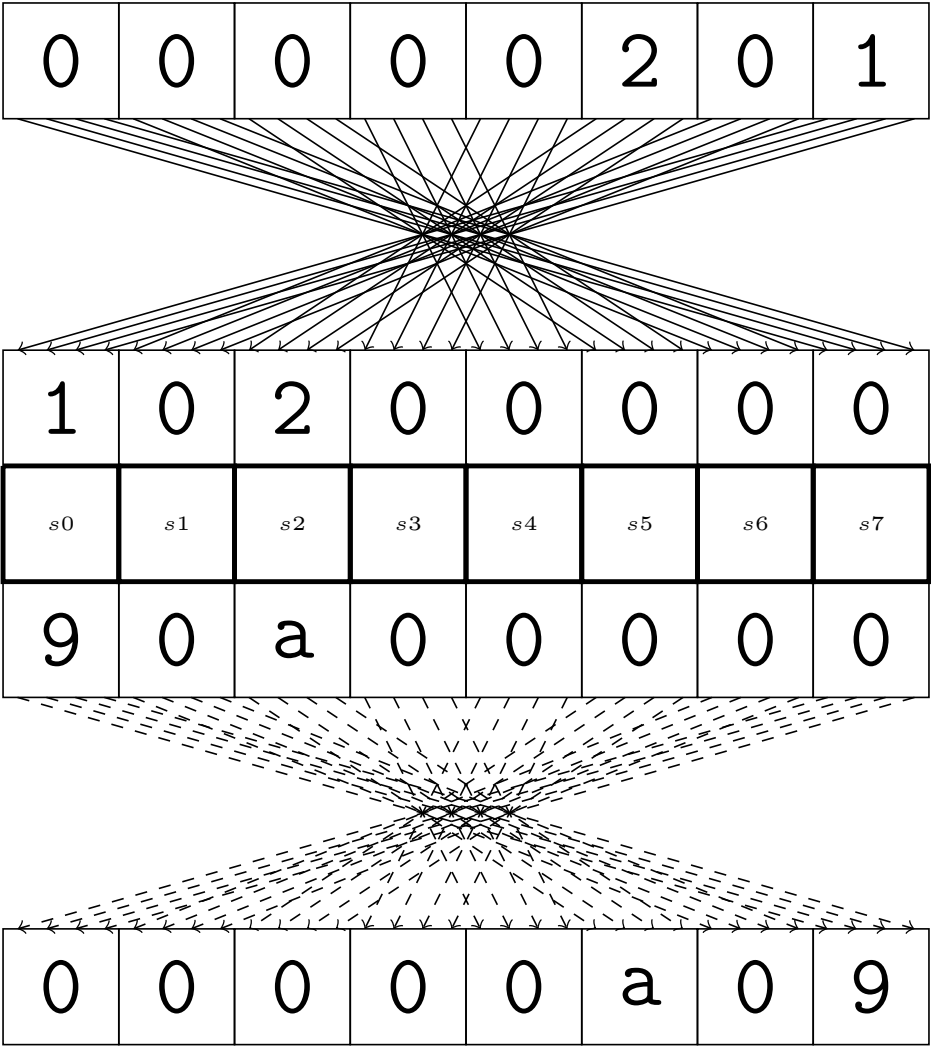
6 LBlock_round

Maximal linear correlation: $2^{-2.0}$



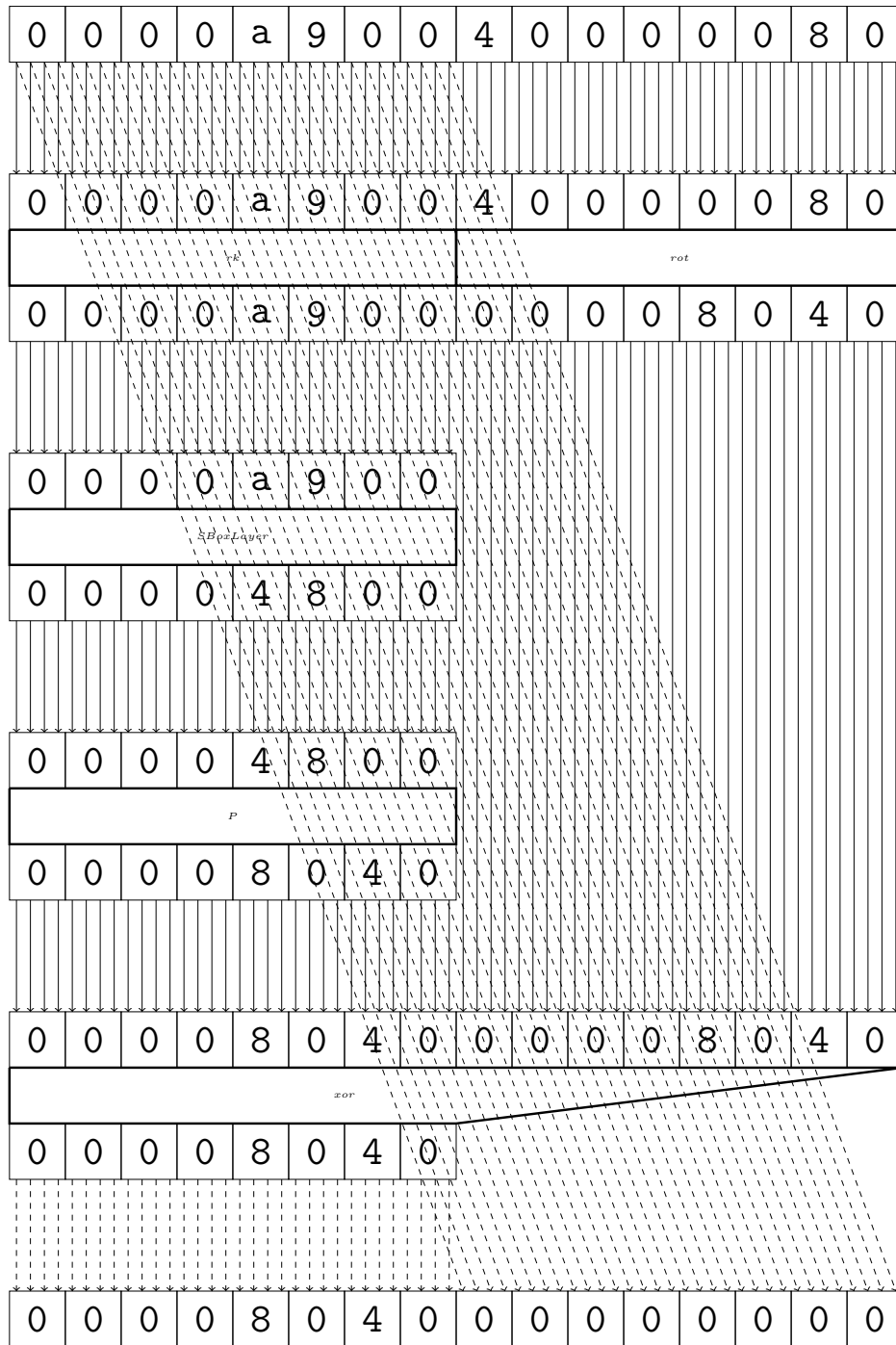
7 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



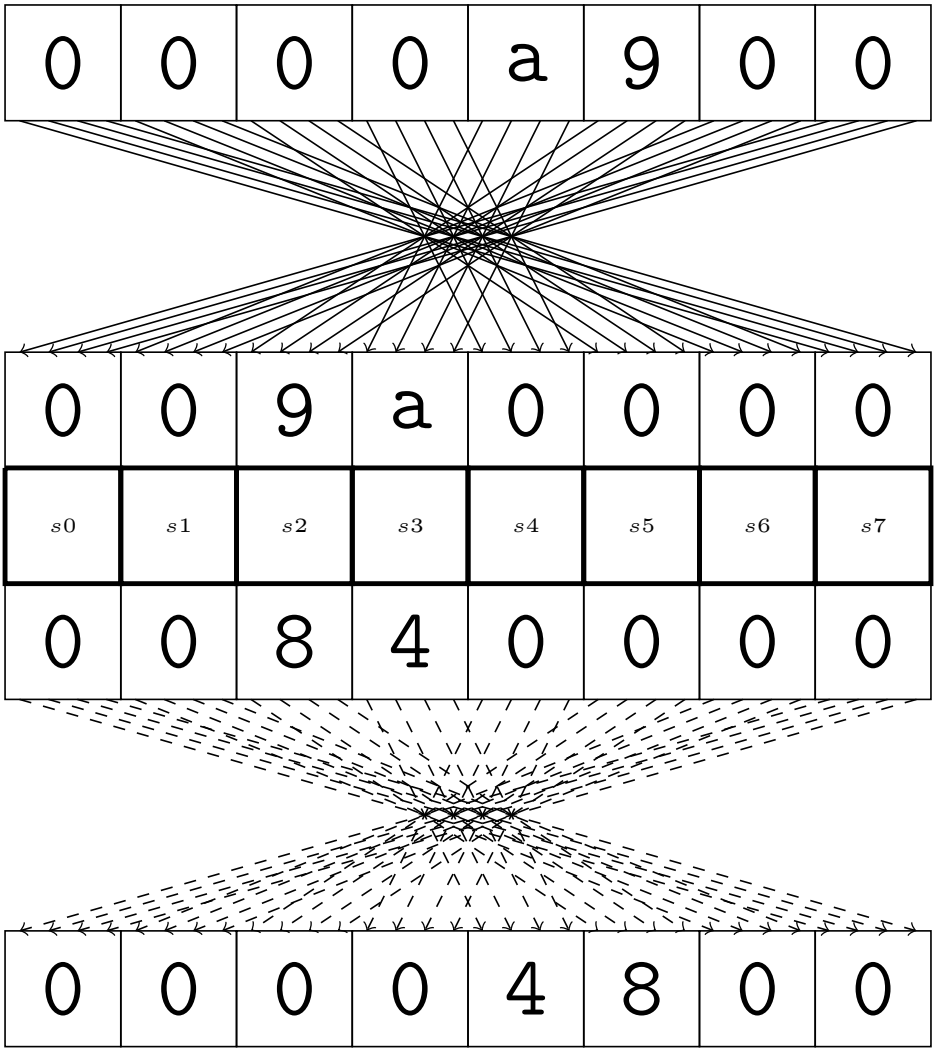
8 LBlock_round

Maximal linear correlation: $2^{-2.0}$

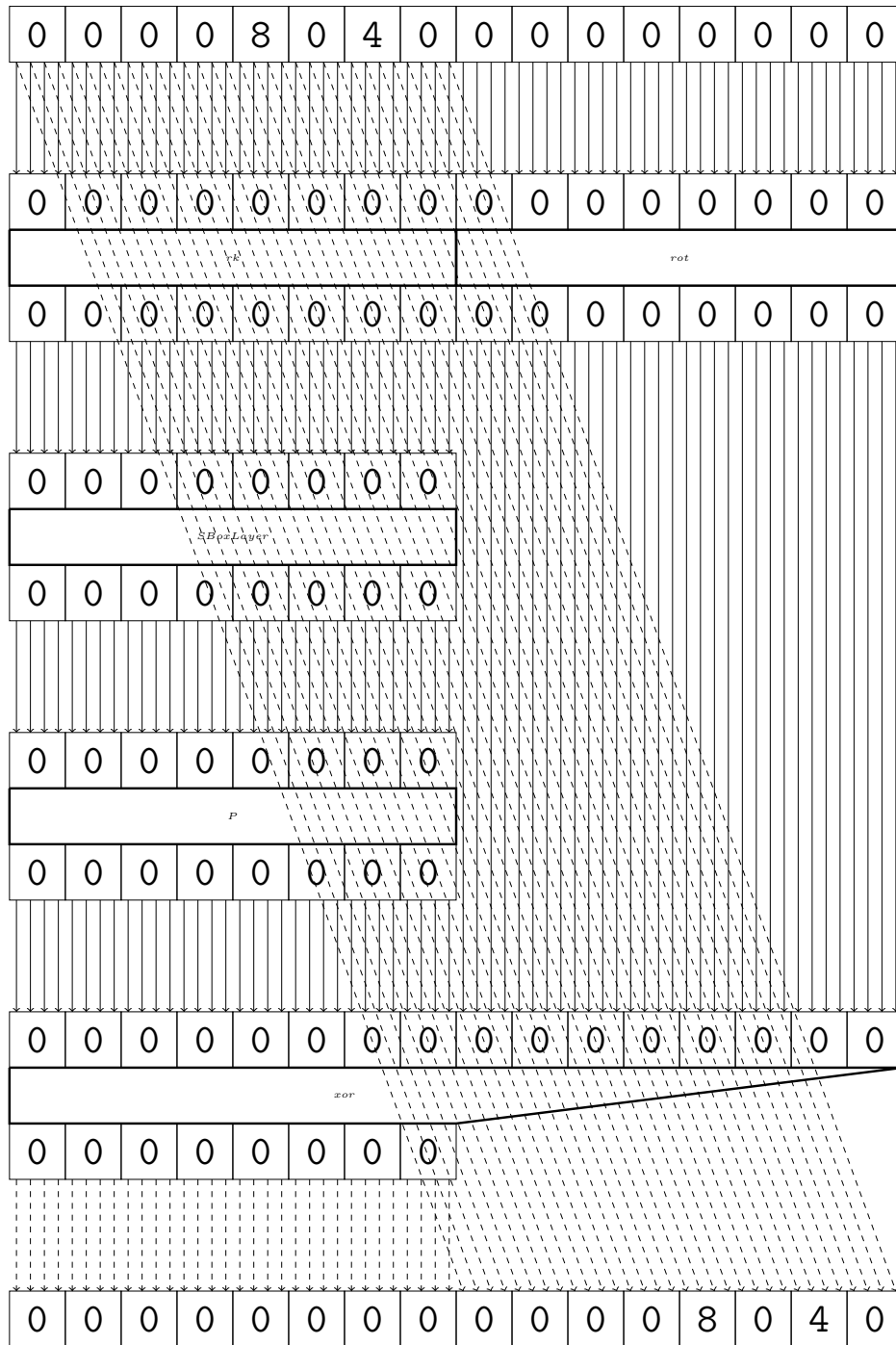


9 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

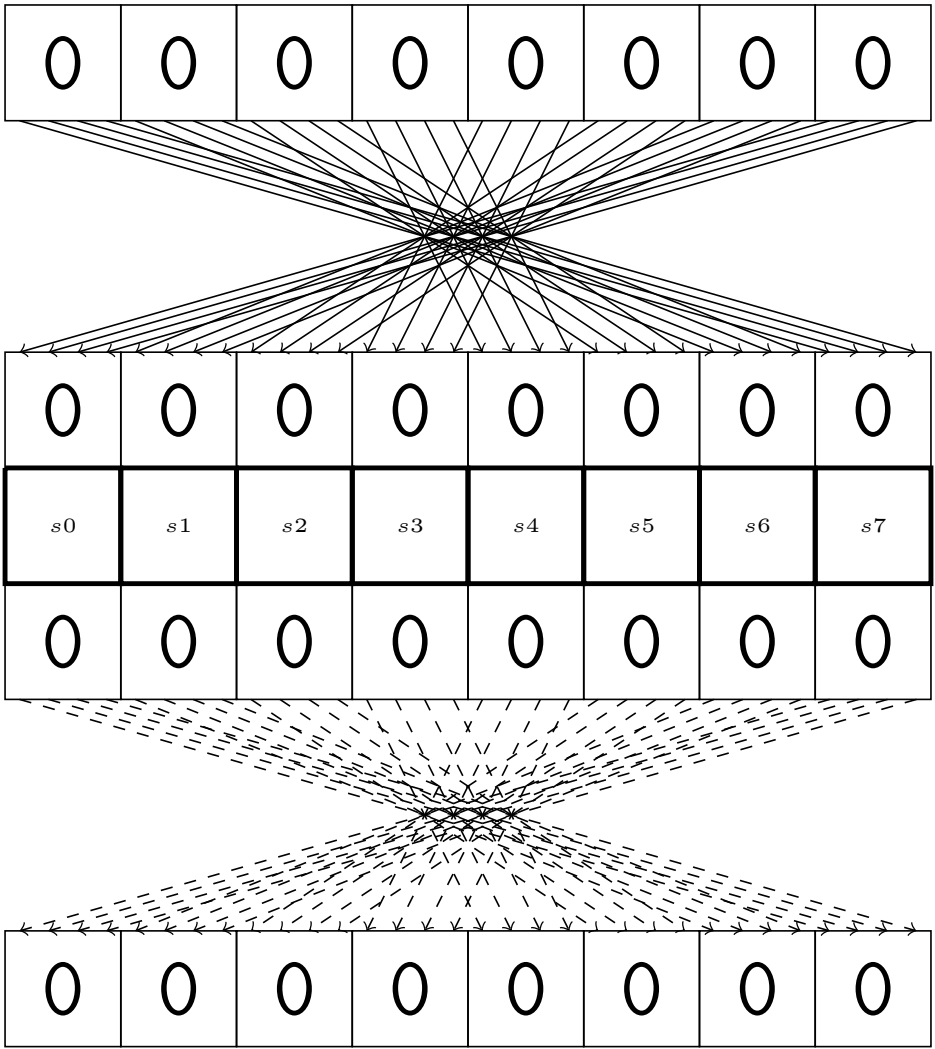


10 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

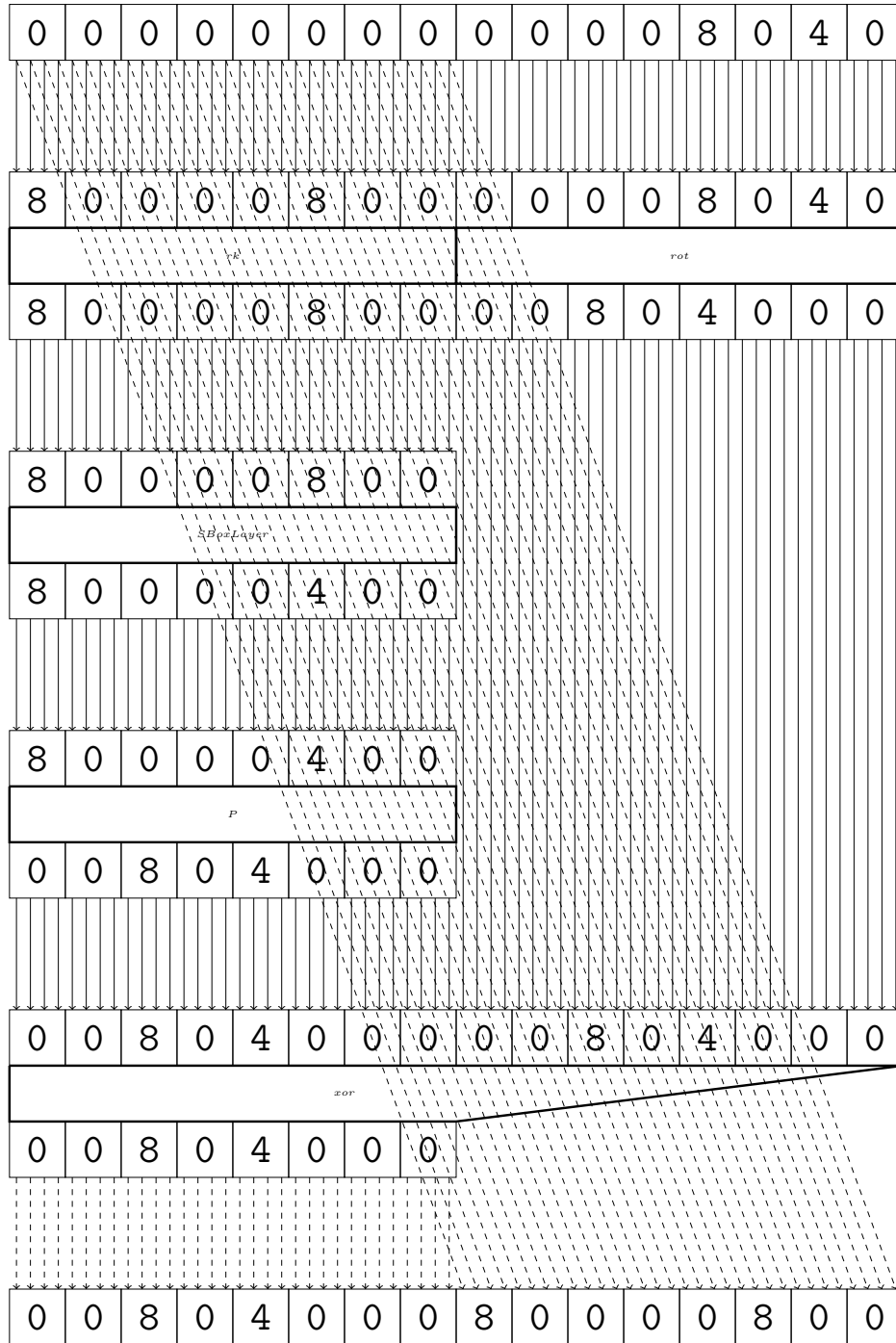
11 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



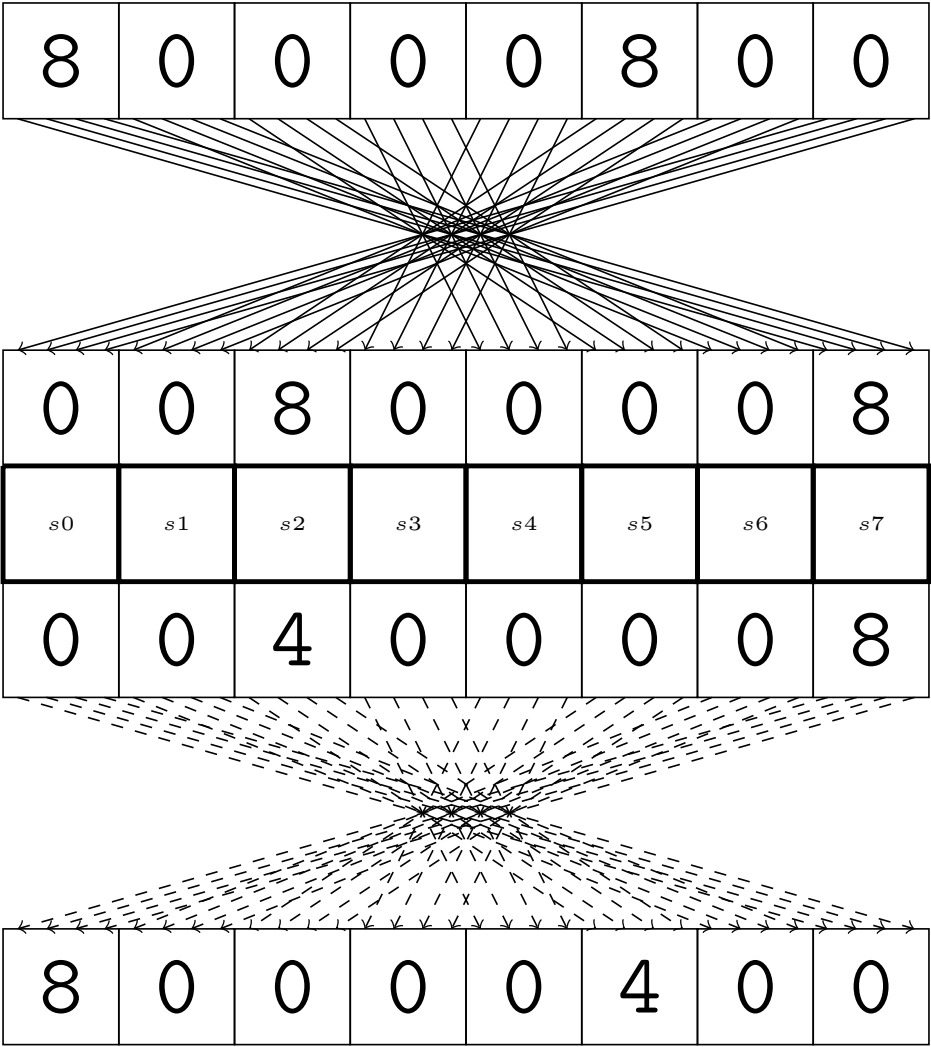
12 LBlock_round

Maximal linear correlation: $2^{-2.0}$



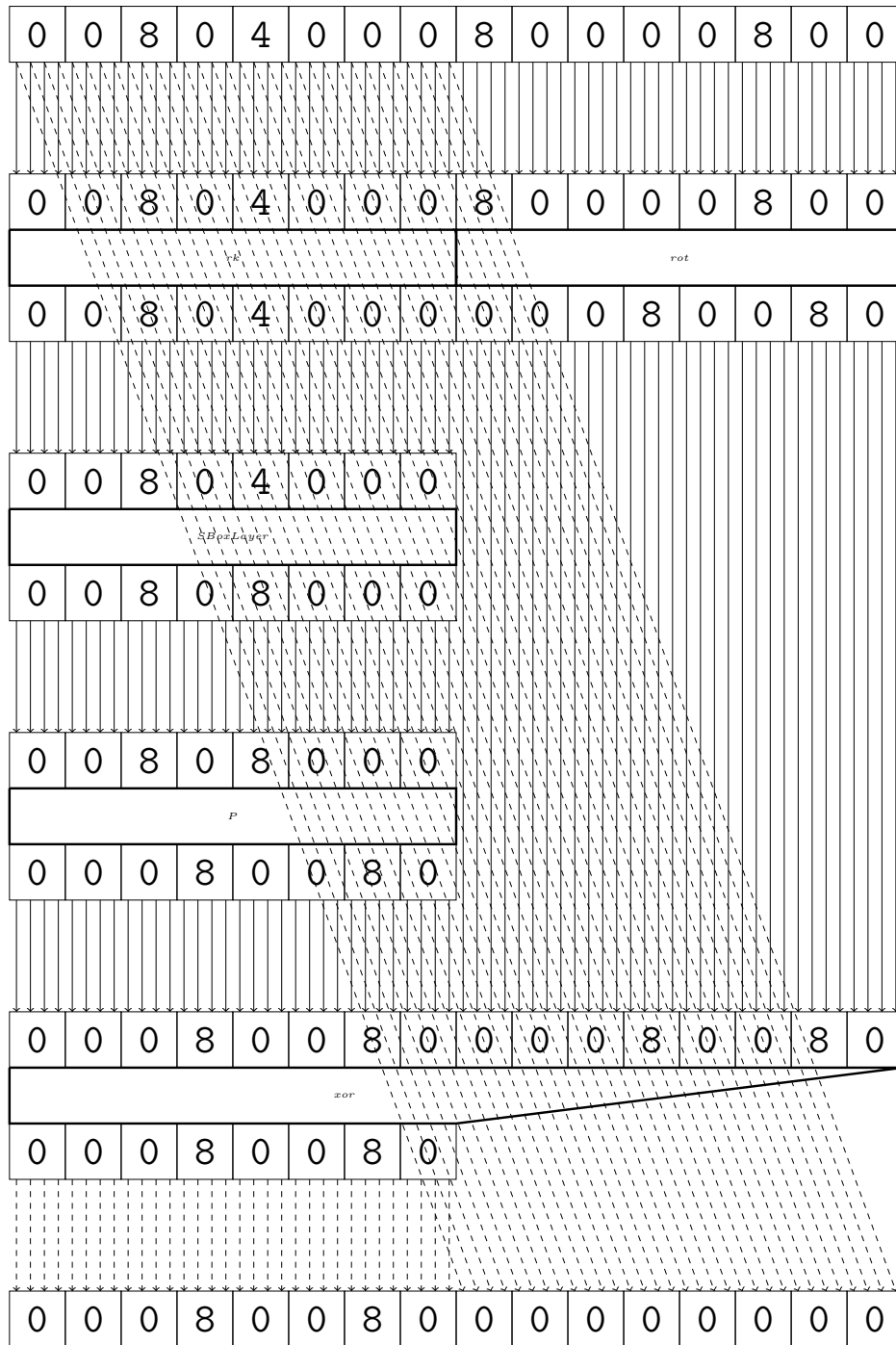
13 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



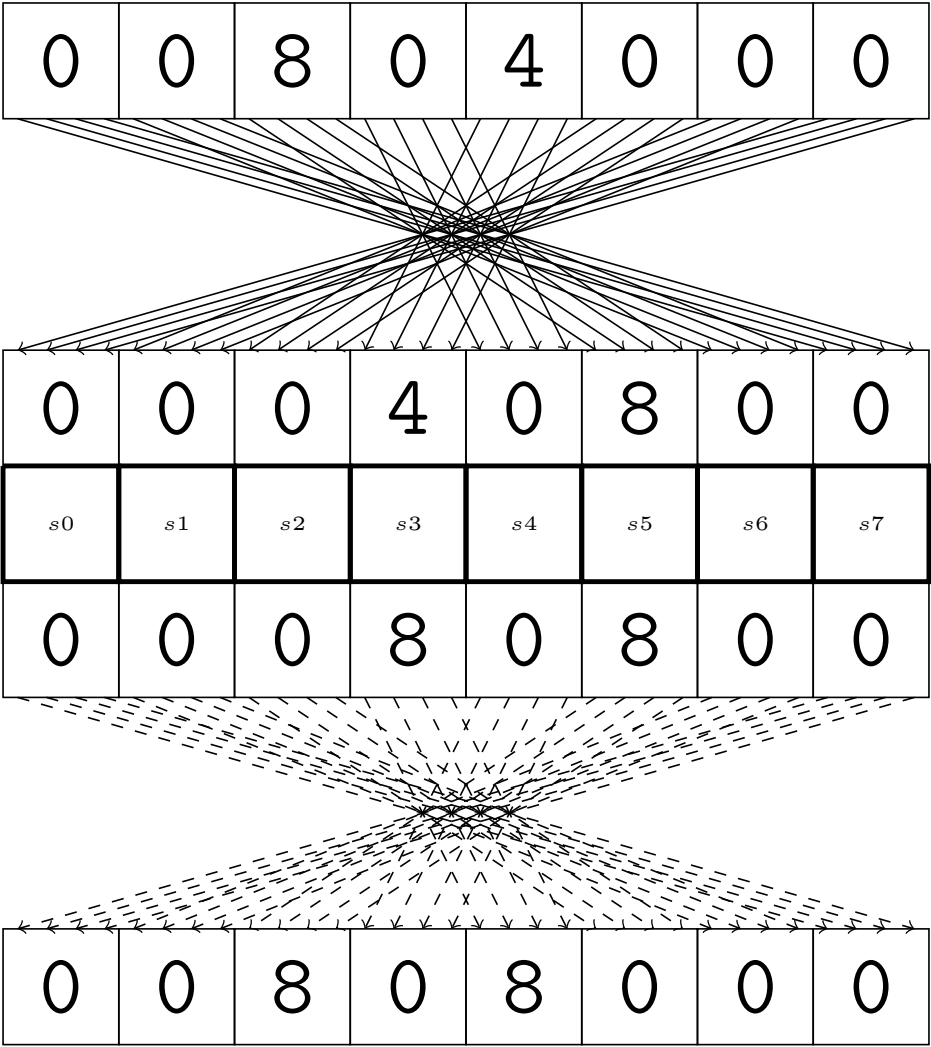
14 LBlock_round

Maximal linear correlation: $2^{-2.0}$



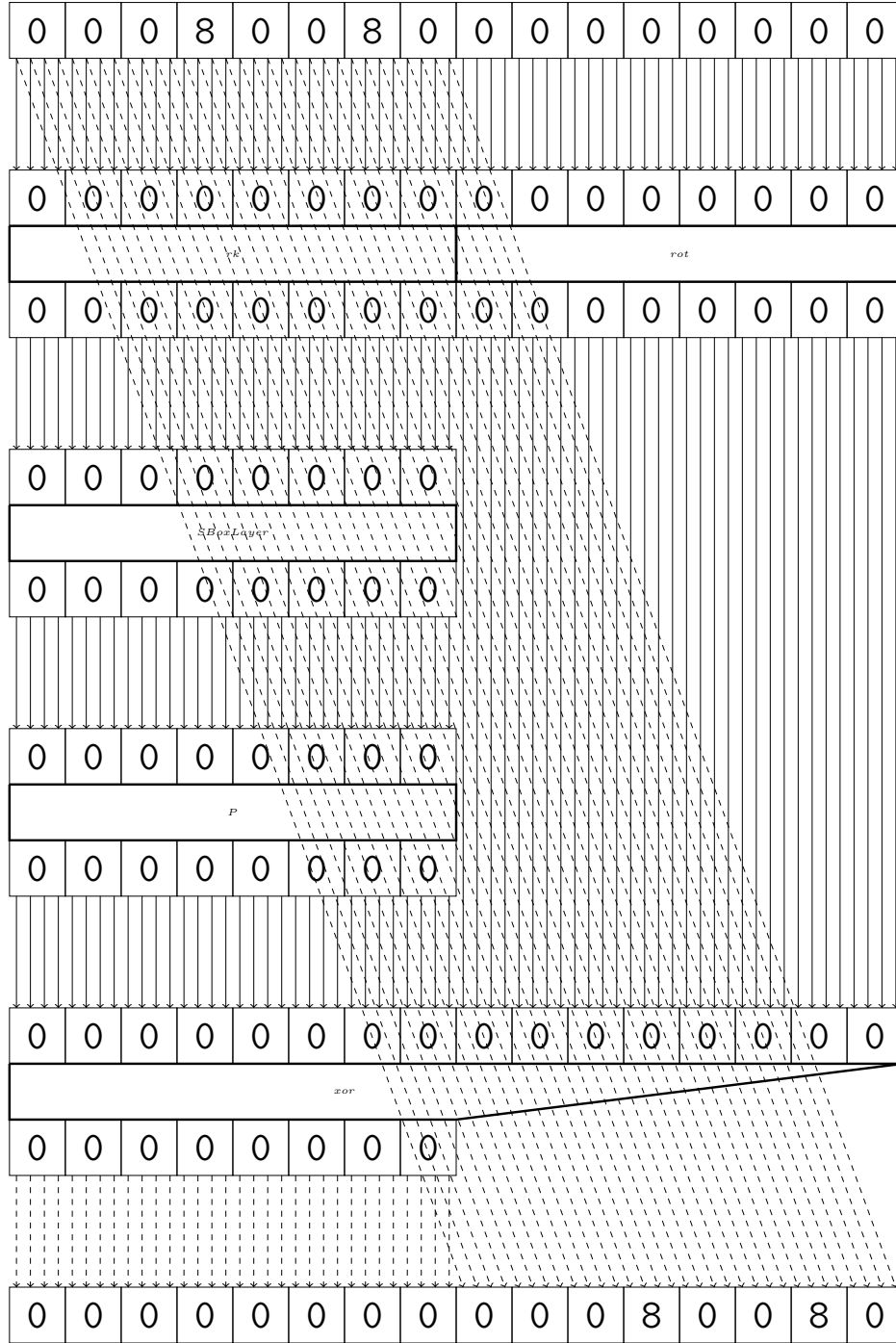
15 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



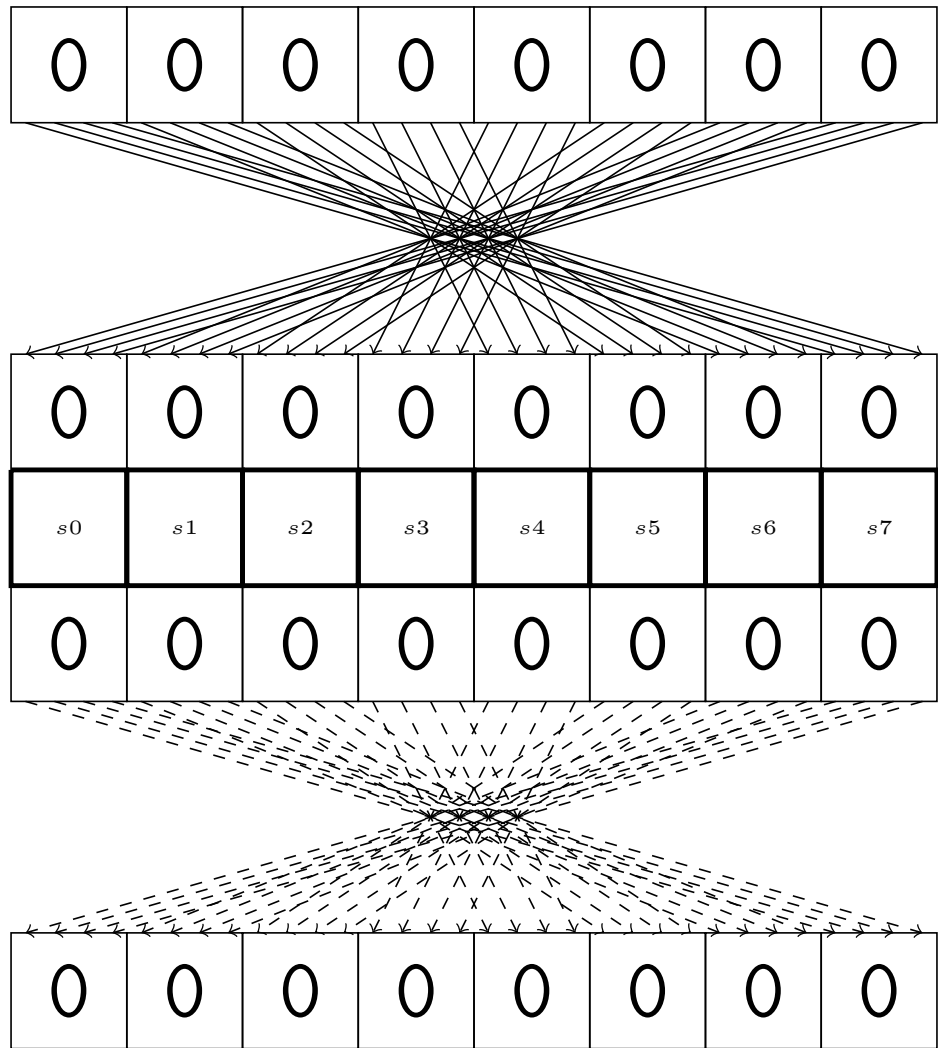
16 LBlock_round

Maximal linear correlation: $2^{-0.0}$



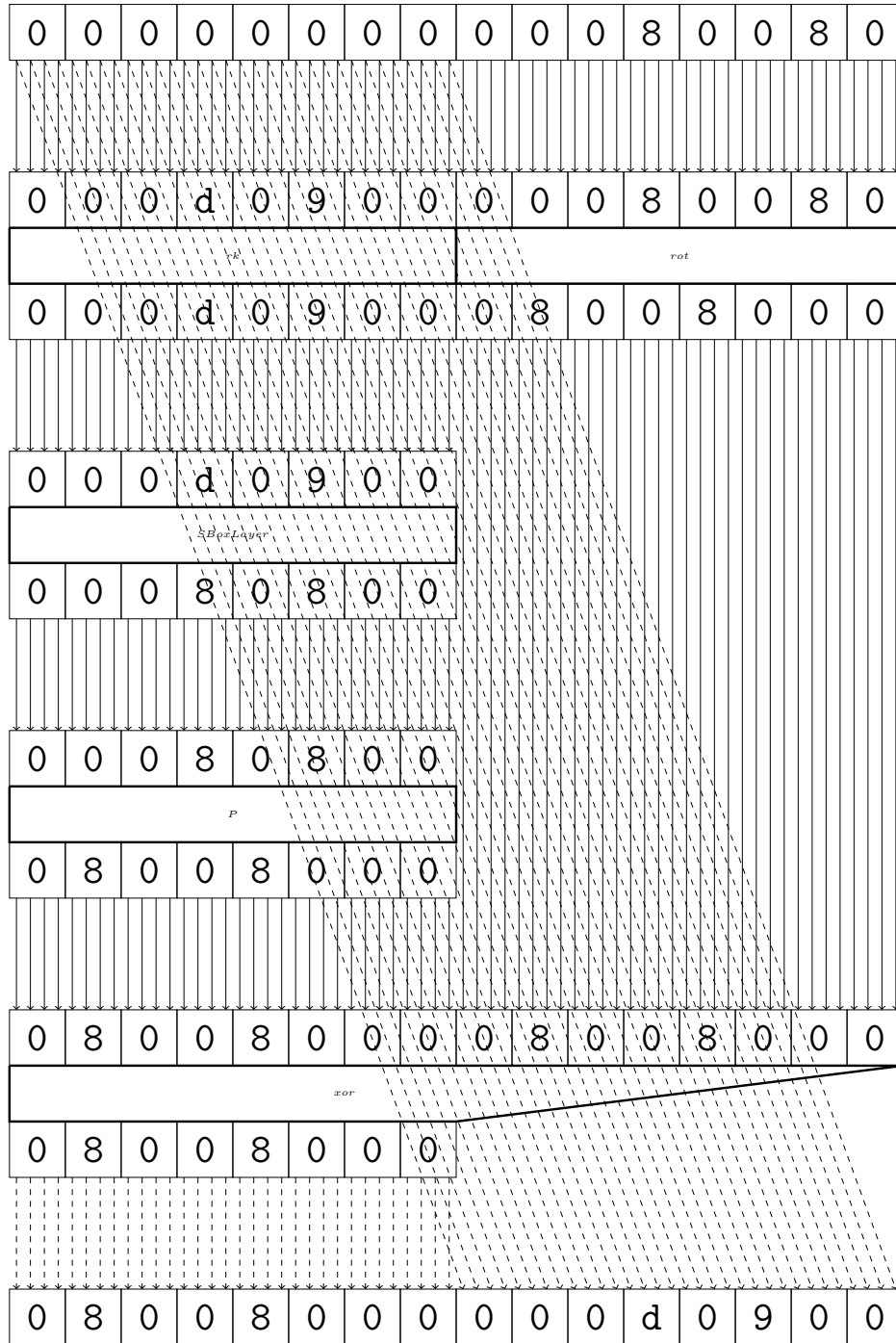
17 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



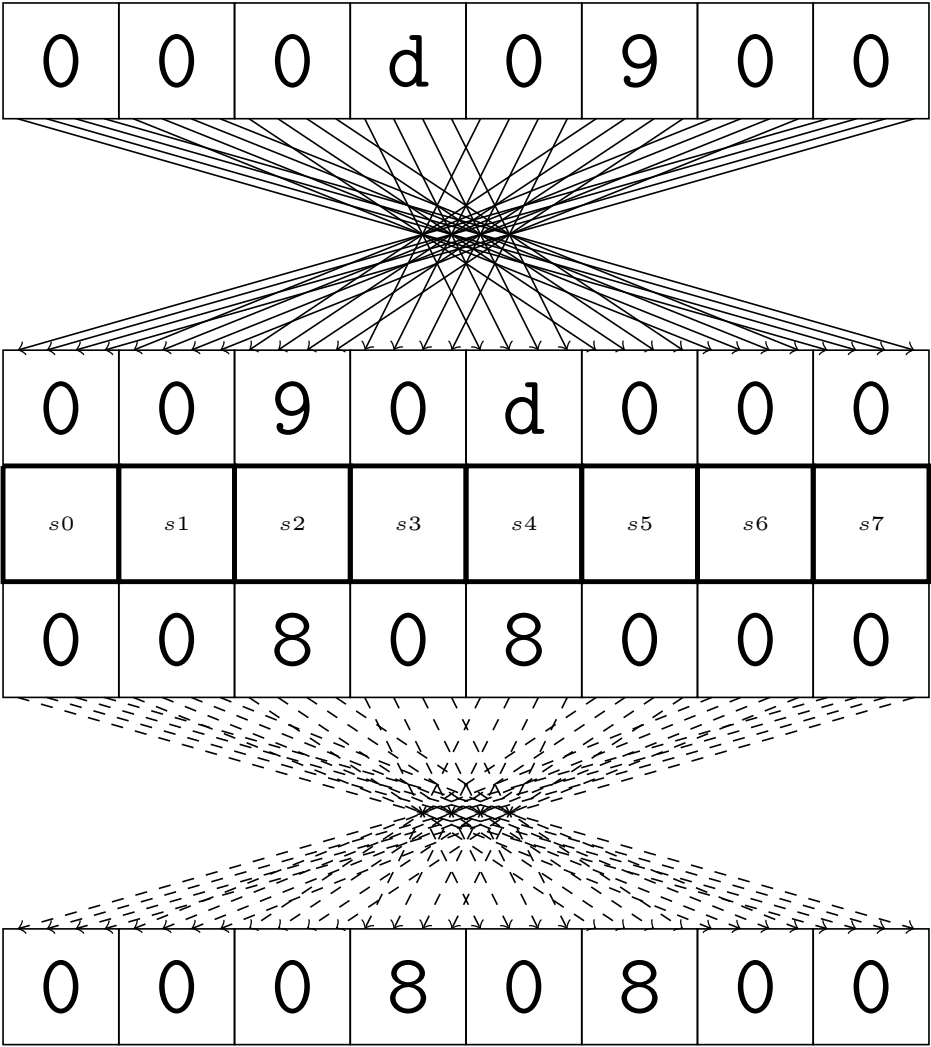
18 LBlock_round

Maximal linear correlation: $2^{-2.0}$

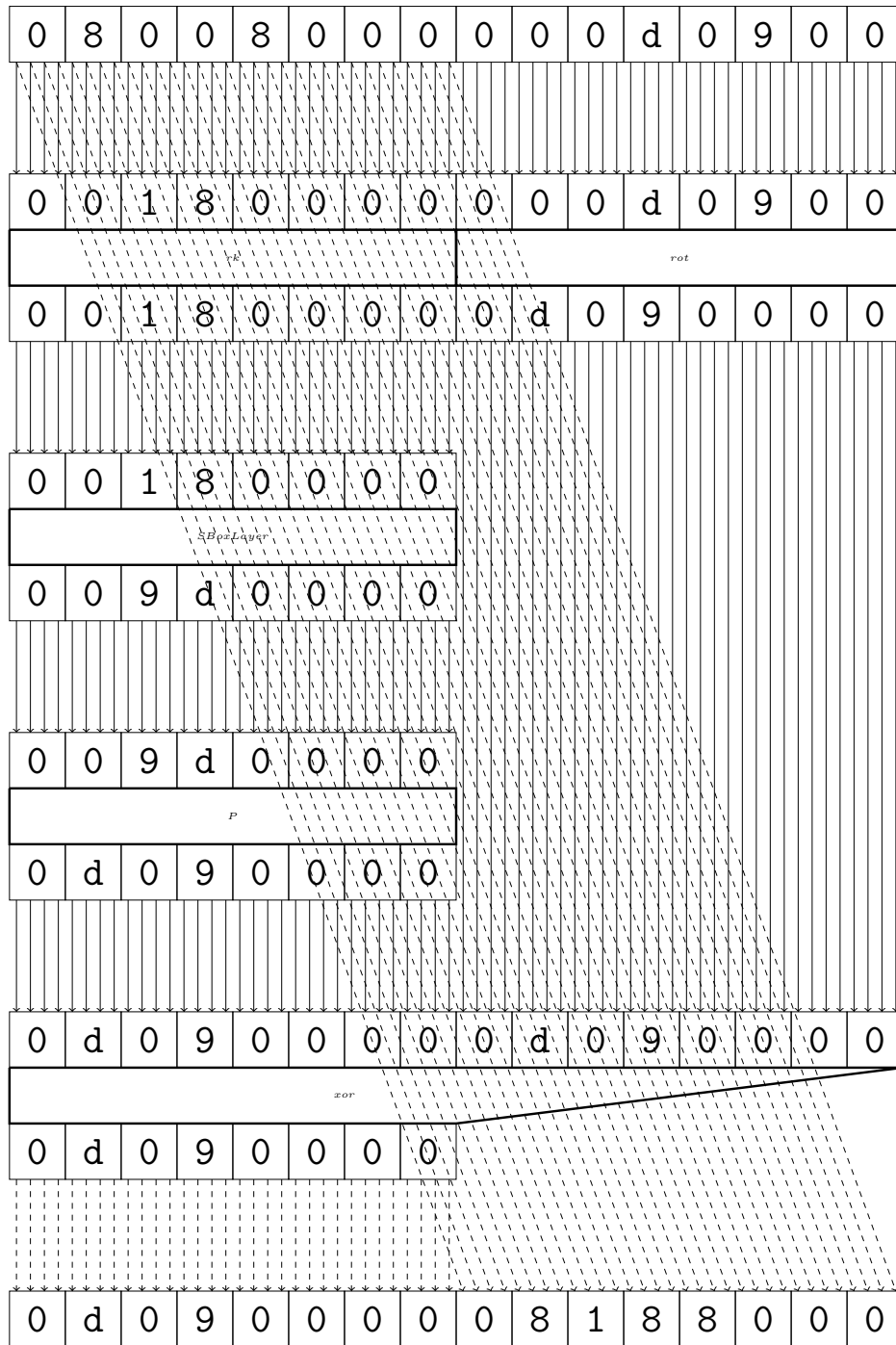


19 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

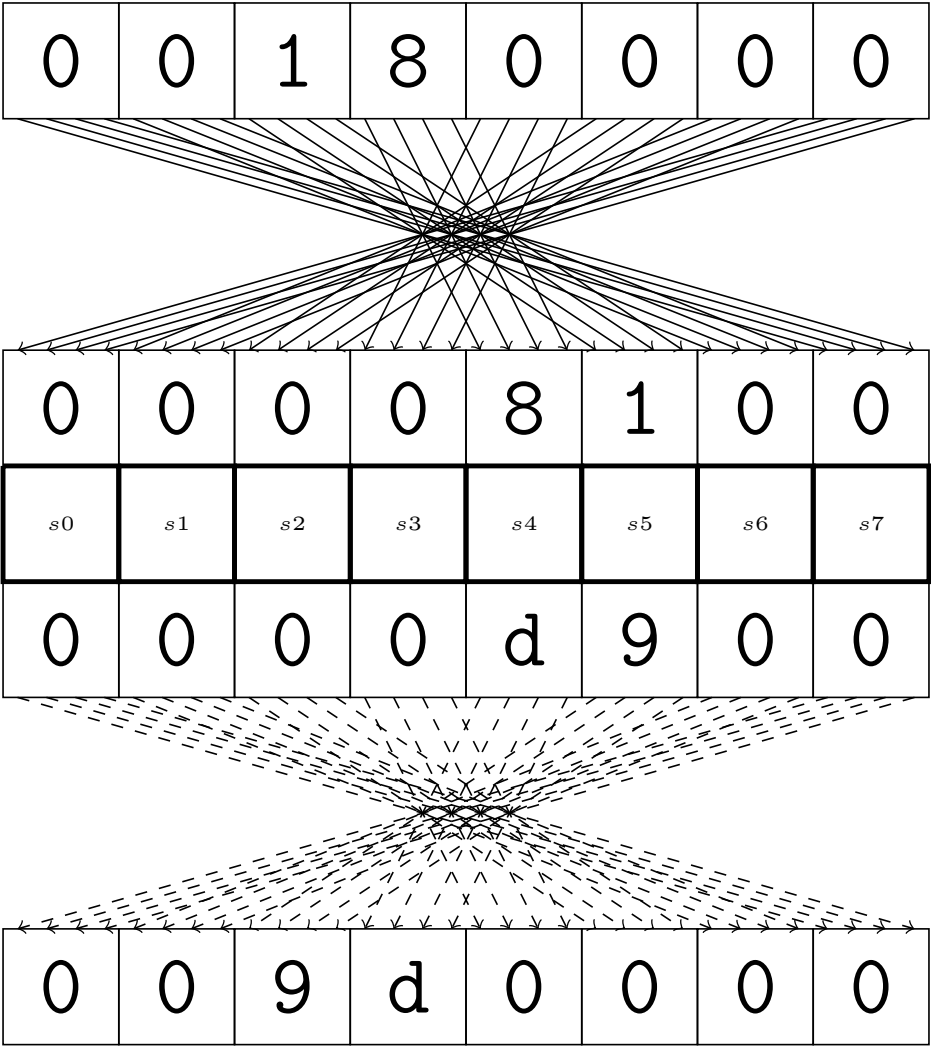


20 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

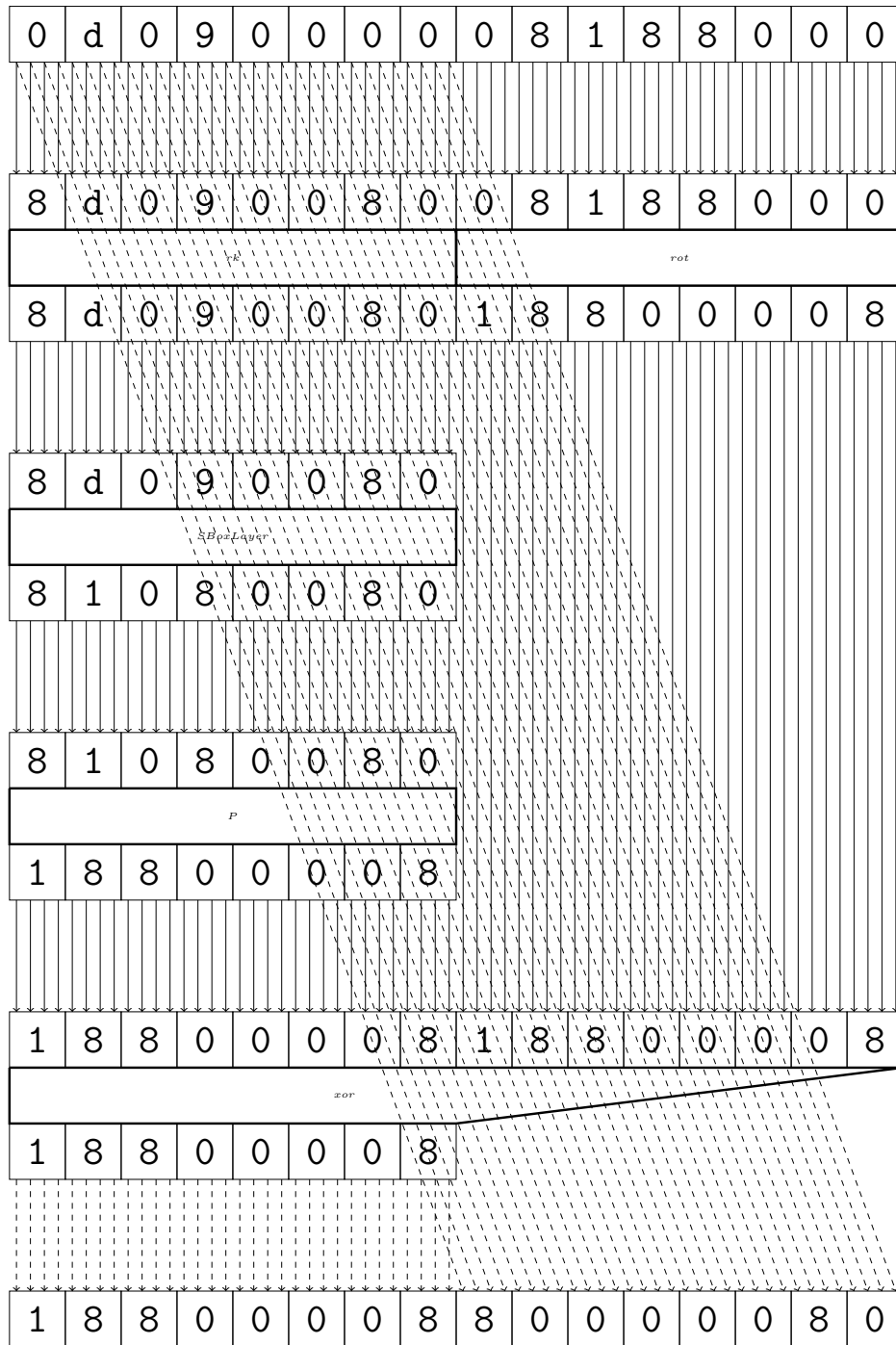
21 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



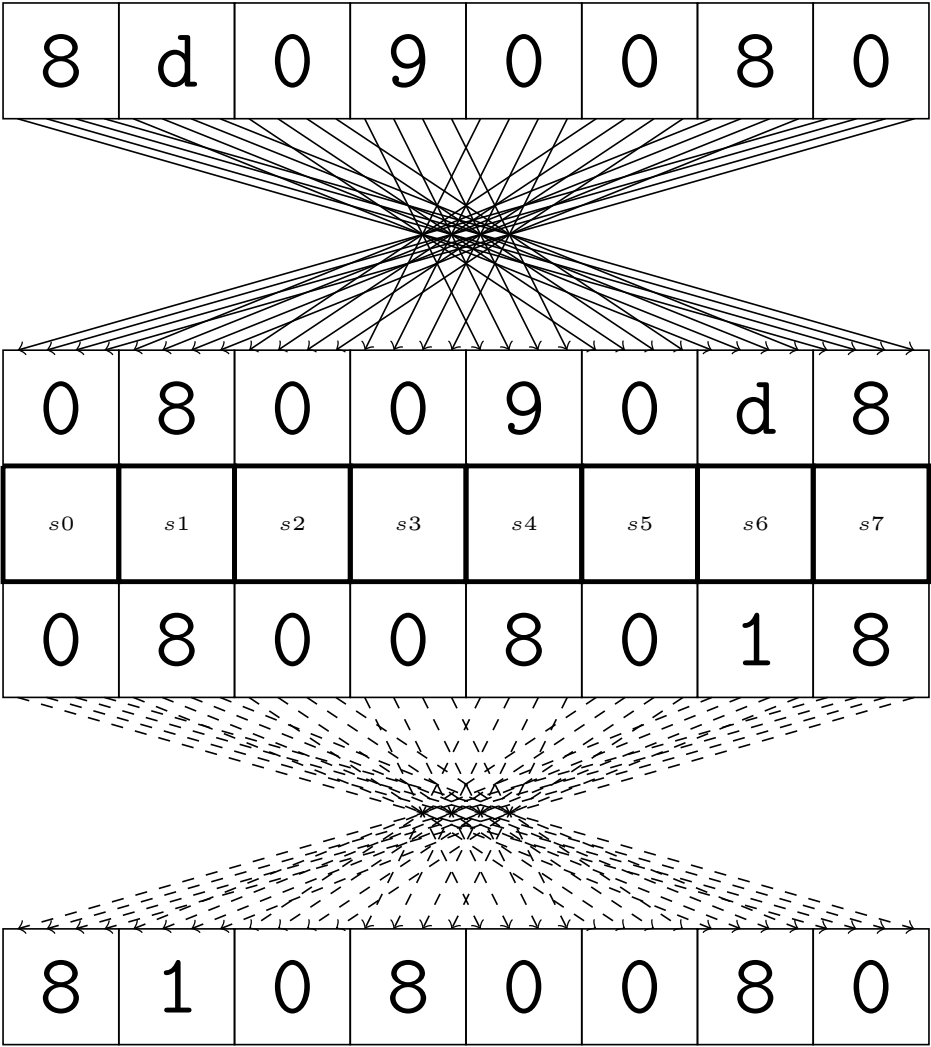
22 LBlock_round

Maximal linear correlation: $2^{-4.0}$



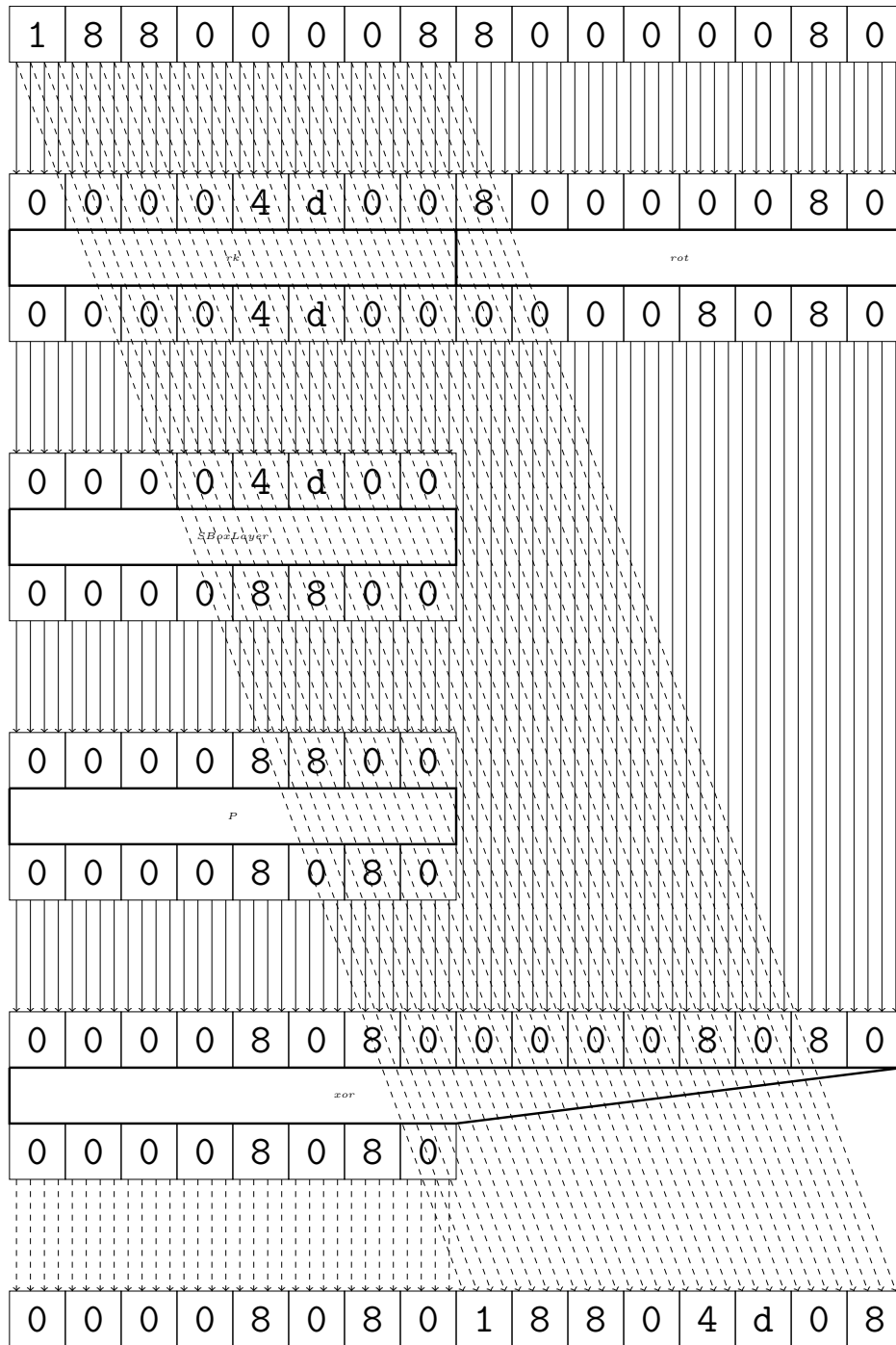
23 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



24 LBlock_round

Maximal linear correlation: $2^{-2.0}$



25 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

